

A. MEASUREMENT AND BIG PICTURE

A.1 Macro (L1)

Aggregates $C_t = \sum_i c_{it}$, general equilibrium, dynamic, quantitative. History: Keynes, IS-LM (Hicks), Okun (-2% GDP $\leftrightarrow$ $+1$ pt u), Phillips menu, Friedman-Phelps vertical LR PC, 70s stagflation, Lucas critique, RE, Kydland-Prescott RBC, New Keynesian (sticky prices $\Rightarrow$ money non-neutral), VARs $\mathbf{Y}_t = \alpha + B\mathbf{Y}_{t-1} + u_t$.

Lucas critique. $Y_t = \alpha + \delta(\tau)T_t(\tau) + \varepsilon_t$, $T_t = \tau Y_t$. Reduced-form δ moves with policy τ , so old $\hat{\delta}$ gives wrong counterfactuals. Fix: model preferences/technology.

A.2 National accounts (D1)

GDP = value of *final* goods produced in a period. Three ways: value added $\sum[\text{output} - \text{intermediates}]$ (inventory counts); expenditure $Y = C + I + G + NX$ (G excludes transfers, interest); income = factor payments + indirect taxes. $NI = GDP + NFP - D - NIT$. (B) $Y + TR + NFP + NTR = C + T + S$. (C) $T + \dot{M} + \dot{B}^G = G + TR$. (D) $X + NFP + NTR + CF = IM$.

(E) Sub (A) into (B), cancel C , use (D): $I = S + (T - G - TR) + CF$. Closed, no gov: $S = I$.

CPI overstates cost of living: quality bias, substitution bias.

Examples: French tourist eats US burger $\Rightarrow$ US $NX \uparrow$, $Y \uparrow$; French $C \uparrow$, $IM \uparrow$, Y same. Car made 2008 sold 2009: 2008 $I \uparrow$, 2009 $C \uparrow$, $I \downarrow$.

B. ENDOWMENT ECONOMY (L2, D2, PS2)

B.1 Setup

$t = 0..T$ (T odd), types $e \in \{A, B\}$, masses n^A, n^B , A gets $1, 0, 1, 0, \dots$, B gets $0, 1, 0, 1, \dots$ $U = \sum \beta^t \ln c_t$, $\beta = 1/(1+\rho)$.

Nominal to real. $p_t c_t + s_t \leq p_t y_t + (1 + i_{t-1})s_{t-1}$. Divide by p_t , $b_t = s_t/p_t$, $1 + \pi_t = p_t/p_{t-1}$, $1 + r_{t-1} = \frac{1+i_{t-1}}{1+\pi_t}$: $c_t + b_t \leq y_t + (1 + r_{t-1})b_{t-1}$, $b_{-1} = 0$. Fisher $i_t \approx r_t + \pi_{t+1}$.

Binds since $u' > 0$: slack $\Rightarrow$ raise c_t , contradiction. **NPG (finite)** $b_T \geq 0$.

B.2 Competitive equilibrium

Prices $\{r_t\}$ and allocations $\{c_t^A, c_t^B, b_t^A, b_t^B\}$ s.t. (1) taking $\{r_t\}$ as given, allocations solve HH problems; (2) goods: $n^A c_t^A + n^B c_t^B = n^A y_t^A + n^B y_t^B \forall t$; (3) bonds: $n^A b_t^A + n^B b_t^B = 0 \forall t$.

B.3 KKT

$\mathcal{L} = \sum \beta^t [\ln c_t + \lambda_t (y_t + (1 + r_{t-1})b_{t-1} - c_t - b_t)] + \mu_T b_T$.

(1) $1/c_t = \lambda_t$. (2) $\lambda_t = \beta(1 + r_t)\lambda_{t+1}$, $t < T$. (3) $\mu_T = \beta^T \lambda_T$. (4)-(5) constraints, (6)-(7)

slackness, (8)-(9) $\lambda, \mu \geq 0$. $\lambda > 0 \Rightarrow$ budget binds, $\mu_T > 0 \Rightarrow b_T = 0$. $\frac{1}{c_t} = \beta(1 + r_t) \frac{1}{c_{t+1}}$

IBC. $q_t = 1/\prod_{\tau < t}(1 + r_\tau)$, multiply budget by q_t , telescope: $\sum q_t c_t = \sum q_t y_t$.

B.4 Equilibrium

Goods clearing + Euler: $Y_t = \frac{1}{\beta(1+r_t)} Y_{t+1}$, so $1 + r_t = \frac{1}{\beta} \frac{Y_{t+1}}{Y_t}$. Alternating: $1 + r_t = \frac{n^B}{\beta n^A}$

(t even), $\frac{n^A}{\beta n^B}$ (odd). Scarce today $\Rightarrow$ high rate.

$n = n^B/n^A$: $q_t = \beta^t$ (even), β^t/n (odd). $\sum_{\text{even}} \beta^t / \sum \beta^t = \frac{1}{1+\beta}$.

$c_{ev}^A = \frac{1}{1+\beta}$, $c_{od}^A = \frac{n}{1+\beta}$, $c_{ev}^B = \frac{\beta}{n(1+\beta)}$, $c_{od}^B = \frac{\beta}{1+\beta}$. $n = 1$: flat, A eats more (endowment first).

Two-period (D2). $q = 1/(1+r)$, $c_1 + q c_2 = y_1 + q y_2$. $c_1 = \frac{y_1 + q y_2}{1+\beta}$, $c_2 = \frac{\beta}{q} c_1$, $b = \frac{\beta y_1 - q y_2}{1+\beta}$.

Clearing: $q = \beta Y_1/Y_2$. Ex: $(3, 1), (1, 1)$, $\beta = .9$: $q = 1.8$, $c^A = (2.526, 1.263)$, $b^A = .474$.

C. INFINITE HORIZON, UNCERTAINTY

C.1 NPG and TVC

Ponzi: $b_t = -x \prod_{\tau < t}(1 + r_\tau)$ rolls debt forever.

NPG: $\lim_T \frac{b_T}{\prod_{\tau=0}^{T-1}(1 + r_\tau)} \geq 0$. TVC: $\lim \beta^T \lambda_T b_T = 0$. Iterate (2): $\lambda_t = \frac{\lambda_0}{\beta^t \prod_{\tau < t}(1 + r_\tau)}$, plug

in, divide by λ_0 : NPG with =.

C.2 Stochastic Euler

$y_t = \bar{y} + \varepsilon_t$. FOC b_t : $\lambda_t = \beta(1 + r_t)E_t \lambda_{t+1}$ (r_t known at t): $u'(c_t) = \beta(1 + r_t)E_t[u'(c_{t+1})]$

C.3 Identical agents (PS6)

Zero net supply $\Rightarrow b = 0$, $c = y$: $R(y) = \frac{u'(y)}{\beta E[u'(y')|y]}$. Log, $\{2, 1\}$, stay prob p :

$R_h = \frac{1}{\beta(2-p)}$, $R_l = \frac{2}{\beta(1+p)}$, $R_h < \frac{1}{\beta} < R_l$. iid $\frac{1}{2}$: $R(2) = \frac{2}{3\beta}$, $R(1) = \frac{4}{3\beta}$. Boom $\Rightarrow$ want to save $\Rightarrow R$ low. Bellman $V(b, y) = \max_{a'} \{u(b + y - a') + \beta E[V(R(y)a', y')|y]\}$, $b = R_{t-1}a_t$.

D. LOG-LINEARIZATION (D3, PS3)

$\hat{x} = \frac{x-\bar{x}}{\bar{x}} \approx \ln x - \ln \bar{x}$, $x \approx \bar{x}e^{\hat{x}}$.

Methods. (1) sub $\bar{x}e^{\hat{x}}$, divide by SS, take logs. (2) Taylor in levels, divide by $\bar{Y}$. (3) take logs, total differential, $dx/\bar{x} = \hat{x}$. $Y = AK^\alpha N^{1-\alpha}$: $\hat{Y} = \hat{A} + \alpha \hat{K} + (1-\alpha)\hat{N}$.

Rules. $cx \rightarrow \hat{x}$. $x^a \rightarrow a\hat{x}$. $xz \rightarrow \hat{x} + \hat{z}$. $x/z \rightarrow \hat{x} - \hat{z}$. $x \pm z \rightarrow \frac{\bar{x}}{\bar{y}}\hat{x} \pm \frac{\bar{z}}{\bar{y}}\hat{z}$.

$Y = C + I$: $\hat{Y} = \frac{\bar{C}}{\bar{Y}}\hat{C} + \frac{\bar{I}}{\bar{Y}}\hat{I}$. $K' = (1-\delta)K + I$: $\hat{K}' = (1-\delta)\hat{K} + \delta\hat{I}$ ($\bar{I} = \delta\bar{K}$). CRRA

Euler: $\hat{C}_{t+1} - \hat{C}_t = \frac{1}{\gamma}\hat{R}_t$.

D.1 Working (1960)

$u_{t+1} = \frac{c_{t+1} + c_t}{2}$ every other period, c random walk. $u_{t+1} = x_{t+1} - x_{t-1} = \frac{1}{2}[\varepsilon_{t+1} + 2\varepsilon_t + \varepsilon_{t-1}]$.

$\text{Var } u = \frac{3}{2}\sigma^2$, $\text{Cov}(u_{t+1}, u_{t-1}) = \frac{1}{4}\sigma^2$, $\rho_1 = \frac{1}{6}$. $\text{Cov}(u_{t+1}, u_{t-3}) = 0$: skip one observation.

E. DYNAMIC PROGRAMMING (L3, D4)

E.1 Problems 1 and 2

P1: $\max \sum \beta^t \hat{F}(y_t)$, $x_{t+1} = h(x_t, y_t) \in \Gamma(x_t)$ (x state, y control). Invert $y_t = \hat{h}(x_t, x_{t+1})$, $F(x, x') = \hat{F}(\hat{h}(x, x'))$. P2: $\max \sum \beta^t F(x_t, x_{t+1})$.

$$\boxed{F_2(x_t, x_{t+1}) + \beta F_1(x_{t+1}, x_{t+2}) = 0} \quad \text{TVC} \lim \beta^T F_1(x_T, x_{T+1}) x_T = 0.$$

Growth. $c + k' = f(k) + (1 - \delta)k$, $\Gamma(k) = [0, f(k) + (1 - \delta)k]$, $F = u(f(k) + (1 - \delta)k - k')$, $F_1 = u'(f' + 1 - \delta)$, $F_2 = -u'$: $u'(c_t) = \beta u'(c_{t+1})(f'(k_{t+1}) + 1 - \delta)$. SS: $k^* = \left(\frac{\alpha}{1/\beta - 1 + \delta}\right)^{1/(1-\alpha)}$; $\delta = 1$: $(\alpha\beta)^{1/(1-\alpha)}$.

E.2 Bellman

$V(x_0) = \max\{F(x_0, x_1) + \beta \sum_{t \geq 1} \beta^{t-1} F\}$, max in steps, reindex inner sum = $V(x_1)$:

$$\boxed{V(x) = \max_{x' \in \Gamma(x)} \{F(x, x') + \beta V(x')\}} \quad g(x) = \arg \max, \quad V(x) = F(x, g(x)) + \beta V(g(x)).$$

Principle of optimality.

Properties. F incr. in 1st arg + Γ monotone $\Rightarrow V$ strictly incr. F jointly strictly concave + Γ convex $\Rightarrow V$ strictly concave, unique g .

E.3 Contraction

Metric: $d = 0$ iff equal, symmetric, triangle. Sup norm $d(f, g) = \sup |f - g|$. Contraction mod $\beta \in [0, 1)$: $d(Tf, Tg) \leq \beta d(f, g)$.

Banach. Complete space + contraction $\Rightarrow$ unique $f^* = Tf^*$, $d(Tf_0, f^*) \leq \beta d(f_0, f^*)$.

Blackwell. Monotone ($f \leq g \Rightarrow Tf \leq Tg$) + discounting ($T(f + k) \leq Tf + \beta k$) $\Rightarrow$ contraction.

Verify: $F + \beta f \leq F + \beta g$ pointwise, max preserves. $T(f + k) = \max\{F + \beta f\} + \beta k$ (constant leaves the max).

VFI: $d(T^n V_0, V) \leq \beta^n d(V_0, V) \rightarrow 0$. Unbounded F (log, k^α): completeness may fail.

E.4 Envelope route (FOC, envelope, shift, sub)

FOC $F_2(x, x') + \beta V'(x') = 0$. Envelope $V'(x) = F_1(x, g(x))$ (bracket = 0 by FOC).

$$\boxed{F_2(x, g(x)) + \beta F_1(g(x), g(g(x))) = 0} \quad \textbf{Stochastic (L3)}. \quad V(a, y) = \max_{a'} \{u(a + y - \frac{a'}{1+r}) + \beta E[V(a', y')|y]\}, \text{ FOC } \frac{u'(c)}{1+r} = \beta E[V_a(a', y')|y], \text{ envelope } V_a = u'(c): u'(c) = \beta(1+r) \sum_{y'} P(y|y') u'(c').$$

E.5 Closed forms (ln, $\delta = 1$, k^α)

Guess $E + F \ln k$: FOC $k' = \frac{\beta F}{1+\beta F} k^\alpha$; match $\ln k$: $F = \alpha(1 + \beta F)$:

$$\boxed{F = \frac{\alpha}{1 - \alpha\beta}, \quad k' = \alpha\beta k^\alpha, \quad c = (1 - \alpha\beta)k^\alpha} \quad E = \frac{1}{1-\beta} \left[\ln(1 - \alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln(\alpha\beta) \right]. \quad \text{Use}$$

$$1 + \beta F = \frac{1}{1 - \alpha\beta}.$$

VFI from 0: $V_1 = \alpha \ln k$, $b_{n+1} = \alpha(1 + \beta b_n)$, $b_n = \alpha \frac{1 - (\alpha\beta)^n}{1 - \alpha\beta}$, saving rate $\frac{\beta b_n}{1 + \beta b_n} \rightarrow \alpha\beta$.

Cake (CRRA γ). $c_{t+1} = \theta c_t$, $\theta = \beta^{1/\gamma}$, $c = (1 - \theta)g$, $V = u(g)(1 - \theta)^{-\gamma}$ (since $\beta\theta^{1-\gamma} = \theta$).

F. NUMERICAL DP (D5, D6, PS4-5)

F.1 VFI on a grid

Grid A with $a_1 = a_{\min}$, $V_0 = 0$, $V_{j+1}(a, y) = \max_{a' \in A} \{u(y + a - \frac{a'}{1+r}) + \beta \sum_{y'} p(y') V_j(a', y')\}$, $c < 0 \Rightarrow -\infty$. Stop at $\max |V_{j+1} - V_j| < \text{tol}$. Approximation, optimization (grid search), integration (weighted sum). Slow but robust.

F.2 AR(1) $z' = (1 - \rho)\bar{z} + \rho z + \sigma\varepsilon$

$$\sigma_z^2 = \frac{\sigma^2}{1 - \rho^2}.$$

Tauchen. N points on $\bar{z} \pm m\sigma_z$, $w = \frac{2m\sigma_z}{N-1}$, $\mu_j = (1 - \rho)\bar{z} + \rho z_j$. $p_{j1} = \Phi\left(\frac{z_1 + w/2 - \mu_j}{\sigma}\right)$, $p_{j\ell} = \Phi\left(\frac{z_\ell + w/2 - \mu_j}{\sigma}\right) - \Phi\left(\frac{z_{\ell-w/2} - \mu_j}{\sigma}\right)$, $p_{jN} = 1 - \Phi\left(\frac{z_N - w/2 - \mu_j}{\sigma}\right)$.

Rouwenhorst. $\bar{z} \pm v$, $v = \sigma_z \sqrt{N-1}$, $P_2 = \begin{pmatrix} p & 1-p \\ 1-q & q \end{pmatrix}$, $P_n = p[P, 0; 0', 0] + (1-p)[0, P; 0, 0'] + (1-q)[0', 0; P, 0] + q[0, 0'; 0, P]$, halve middle rows. $p + q - 1 = \rho$, $p = q = \frac{1+\rho}{2}$. Matches mean, var, autocorr. Use for $\rho > 0.9$. P_3 rows: $(p^2, 2p(1-p), (1-p)^2)$, $(p(1-q), pq + (1-p)(1-q), q(1-p))$, $((1-q)^2, 2q(1-q), q^2)$. Stationary $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ when $p = q$.

G. CONSUMPTION (L4)

G.1 PIH, IES

$c_t + \frac{a_{t+1}}{1+r_t} = y_t + a_t$, NPG. $u'(c_t) = \beta(1 + r_t)u'(c_{t+1})$. $\beta(1 + r) = 1 \Rightarrow c$ flat.

$u = \frac{c^{1-1/\sigma}}{1-1/\sigma}$: $\boxed{\frac{c_{t+1}}{c_t} = [\beta(1 + r_t)]^\sigma}$, $\frac{\partial \ln(c_{t+1}/c_t)}{\partial \ln(1+r)} = \sigma = \text{IES}$, $1/\sigma = \text{RRA}$, $\sigma \in [.1, .5]$. Grows iff $r > \rho$.

G.2 MPC

$c_{t+j} = (R\beta)^{\sigma j} c_t$ into IBC: $\boxed{c_t = (1 - R^{-1}(R\beta)^\sigma)[a_t + \sum_j R^{-j} y_{t+j}]}$ Transitory: $1 - \frac{1}{R} = \frac{r}{1+r} \approx r$. Permanent: $\frac{r}{1+r} \cdot \frac{1+r}{r} = 1$. MPC depends on persistence (Friedman 1957).

G.3 RE errors, Hall

$\varepsilon_{t+1} = s_{t+1} - E_t s_{t+1}$: $E_t \varepsilon = 0$; $E_t[x\varepsilon] = 0$ for $x \in I_t$; regression slope = $E[E_t(\varepsilon x)] / \text{Var } x = 0$.

$u'(c_{t+1}) \approx u'(c_t) + \varepsilon_{t+1}$; Taylor $u'(c_t) \approx u'(c) + u''(c)c(\ln c_t - \ln c)$:

$\boxed{\ln c_{t+1} = \ln c_t + \eta_{t+1}, \quad \eta = \frac{\varepsilon}{cu''(c)}}$ Test $\ln c_{t+1} = \ln c_t + \gamma x_t + \eta$, $H_0 : \gamma = 0$. Shea 95 (union wages) rejects. JPS 2006 rebates \$300-600: $\Delta c = \beta R_{i,t} + \dots$, PIH $\beta = 0$ (anticipated), $\hat{\beta} \approx .2-.3$. Kaplan-Violante 2014.

G.4 Precautionary

$\beta(1 + r) = 1$, 2nd-order Taylor of $u'(c_{t+1})$ at c_t , take E_t : $\boxed{E_t \frac{\Delta c}{c} = -\frac{u'''c}{u''} \cdot \frac{1}{2} E_t \left[\left(\frac{\Delta c}{c} \right)^2 \right]}$ > 0 iff $u''' > 0$ (prudence, Kimball 1990). Quadratic: $u''' = 0$, none. CRRA: prudence $1 + \frac{1}{\sigma}$. Jensen: u' convex $\Rightarrow u'(c_t) = Eu'(c') > u'(Ec') \Rightarrow c_t < E_t c_{t+1}$. Log, sd 5%: $\approx 2 \cdot \frac{1}{2} \cdot .0025 = 0.25\%$.

G.5 Borrowing constraints

$a_{t+1} \geq \underline{a}$. Natural (repay with $c = 0$ forever, $y_{\min}$): $\boxed{a_{t+1} \geq -\frac{1+r}{r} y_{\min}}$

KKT: $\boxed{u'(c_t) = \beta(1 + r)E_t u'(c_{t+1}) + (1 + r)\mu_t}$ $\mu_t \geq 0$, $\mu_t(a_{t+1} - \underline{a}) = 0$. Binding:

$c_t = y_t + a_t - \frac{a}{1+r}$, mpc = 1 (HANK). $\beta(1 + r) = 1$: $u'(c_t) \geq E_t u'(c_{t+1})$, supermartingale, $c, a \rightarrow \infty$ (self-insurance). Quadratic: $c_t \leq E_t c_{t+1}$, isolates the constraint motive from prudence.

Ex: $\beta = 1, r = 0, y = (1, 3)$, no borrowing: $c = (1, 3)$, $\mu = 1 - \frac{1}{3} = \frac{2}{3}$, mpc 1.